NGUYEN GIA NGHI

Ho Chi Minh City, Vietnam | gianghi30032005@gmail.com | 0703540518 | Portfolio | GitHub | LinkedIn

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern | TMA Solutions Jun–Aug 2026

Ho Chi Minh City

- Implemented bug fixes, product improvements, and small features for an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Worked with Mongoose data models and queries, developed and validated API tests, and contributed changes through an Agile/Scrum workflow.

SYSTEMS AND SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine

- Developed a Python pipeline and CLI that can ingest testbed data, validate records against versioned schemas, construct response vectors, and match fingerprints within compatible contexts.

Repository

PIXELATED Studio | Creator platform and self-hosted edge-cloud-gaming research testbed

- Built and released a TypeScript/React creator platform, then extended it into an edge-cloud-gaming testbed that streams interactive workloads from a host owned by users to a lightweight browser client.
- Implemented a Fastify control plane, Electron-based host orchestration, a containerized runtime, GStreamer capture and encoding, and WebRTC streaming, with cross-layer telemetry export.

Repository | Live application | Releases

PIXELATED User | Browser-based WebAssembly gaming platform

- Built and deployed a PWA application that runs NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly libretro cores, without relying on the desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export data including launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser baseline for comparison with the Studio version's edge-cloud-streaming path.

Repository | Live application

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, PWA, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud and DevOps	AWS, GCP, Linux, Docker, Kubernetes, Terraform, GitHub Actions
Systems and media	WebRTC, GStreamer, telemetry instrumentation, latency analysis

EDUCATION

VNUHCM-University of Science Sep 2023–May 2027 (Expected)

Bachelor of Science | Major: Information Technology Ho Chi Minh City

GPA: 8.41/10.00 (3.61/4.00) | 133 credits completed

CERTIFICATIONS AND ENGLISH

- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall – Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2nd October 2025)